



Politecnico di Torino

Qubit Electronics

Lab 5

A. Tudisco, G. Piccinini, F. Riente

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1 Introduction

During this laboratory session, we will focus on analyzing a real double quantum dot device, where confinement is achieved by exploiting Silicon-Silicon Dioxide (Si-SiO₂) materials, by solving the Schrodinger equation.

2 Modify the code

Modify the code of the previous laboratory in order to solve the Schrodinger equation for the device under investigation. The constraints are:

- Number of energy levels to consider is equal to 5
- The tolerance for the convergence is equal to $1e-9$

Question: Report the modified code commenting your modifications.

3 Results

As done in the first lab on QTCAD, perform all analysis on the generated output files.

Question: Report all the results and also make some comments on them. In particular:

- What do the results represent?
- Does the device work as expected?
- In what bonding configuration are the two dots located?